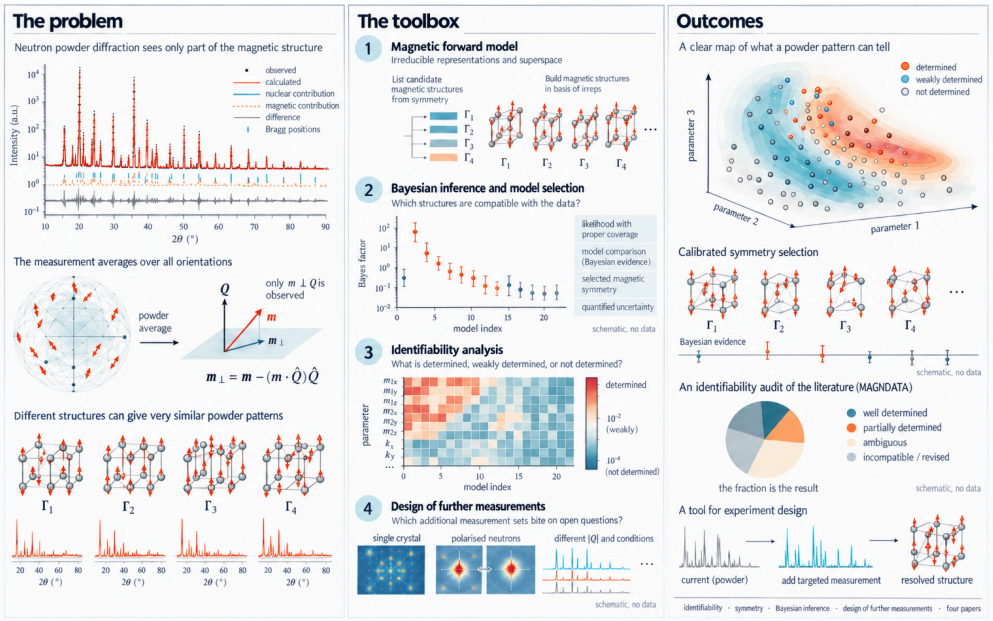
# What a powder pattern can tell

*Identifiability, Bayesian inference and symmetry in the determination of magnetic structures*


## ABSTRACT

More than two thousand magnetic structures are published, most from neutron powder diffraction, and the database collecting them reports that many were described ambiguously; even NiO and MnO were revised in 2024. The measurement is the reason: the neutron sees only the component of the moment perpendicular to the scattering vector, and the powder average discards orientation, so part of a structure is not in the data, part only weakly, and part of what is reported is the model's choice. Symmetry tools list the candidates; the choice is still made by agreement factors and judgement. This thesis asks which features of a magnetic structure a powder pattern determines, which it does not, and what cuts the rest, for one pattern and for the literature. Four contributions: a magnetic forward model in irreducible-representation coordinates, in superspace, that prints the directions the data do and do not fix; magnetic symmetry chosen by Bayesian evidence with a likelihood whose coverage is tested; an identifiability audit of the commensurate structures of MAGNDATA; and which further measurement settles what the powder leaves open. Expected outcomes: a diagnostic in an open engine, calibrated symmetry selection on the group's data, the identifiability map of the literature, a design tool for beam time, four papers.


**Graphical abstract.** Every plot is schematic and says so; the fraction in the audit is the result, not a value assumed.

## REQUIREMENTS

*Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.*

- Bachelor's degree in Physics, Chemistry, Materials Engineering or a related field, and a Master's degree in Physics, Materials Science, Nanotechnology or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in solid-state physics, magnetism, crystallography or diffraction; having used a refinement program (FullProf, GSAS-II or Jana), or some statistics; some programming, ideally in Python.
- Curiosity about what a measurement can and cannot tell, interest in experiments at large neutron facilities, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

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